

Food and Drug Administration



Office of The Commissioner (OC)

FDA Sentinel 3.0 Data Broker Services

DRAFT Performance Work Statement (PWS)

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Background

The Food and Drug Administration Amendments Act of 2007 (FDAAA) directed the U.S. Food and Drug Administration (FDA) to strengthen its postmarket safety surveillance capabilities by establishing an active risk identification and analysis system capable of monitoring the safety of FDA-regulated medical products across a population of at least 100 million individuals. FDAAA further requires FDA to determine whether available postmarket surveillance data and analytic capabilities are sufficient to assess safety concerns prior to requiring additional postmarket studies or clinical trials.

In response, FDA established the Sentinel Initiative—an agency-wide effort to leverage real-world healthcare data and advanced analytic methods to support regulatory decision-making. Since 2009, Sentinel has evolved through multiple phases, including Mini-Sentinel and Sentinel System versions 1.0 and 2.0, expanding FDA's ability to access healthcare data, conduct active surveillance, and respond to emerging public health needs.

FDA Centers have developed complementary postmarket surveillance approaches aligned with their regulated products and statutory responsibilities. CDER implemented the Sentinel System and its Active Risk Identification and Analysis (ARIA) capability. CBER advanced biologics surveillance through the Biologics Effectiveness and Safety (BEST) Initiative. CDRH invested in medical device surveillance capabilities, including collaborations such as the National Evaluation System for Health Technology (NEST), with a continued focus on improving device identification in healthcare data (including Unique Device Identifier (UDI) capture).

Building on these efforts, FDA is transitioning to Sentinel 3.0—a modernized, enterprise approach intended to improve scalability, timeliness, and responsiveness across FDA's postmarket safety surveillance activities. Sentinel 3.0 is structured as coordinated components that include (1) data access and acquisition, (2) a Government-developed data platform for ingestion and analytics enablement, and (3) scientific services to support study execution. This Performance Work Statement (PWS) addresses the data access and acquisition component only.

Purpose

The Government is transitioning Sentinel to a real-time surveillance platform. The goal of this Sentinel 3.0 Data Broker Services is to continue meeting or exceeding legislative requirements, promote program efficiency, and provide governed, scalable access to the data assets needed to support Sentinel 3.0 platform operations and study execution—while maintaining flexibility to address emerging safety issues and rapidly changing public health needs.

The Sentinel 3.0 Initiative fulfills the Food and Drug Administration Amendments Act (FDAAA) of 2007 (Sec 905) requirement for an active postmarket risk identification and analysis system. FDAAA requires that FDA access disparate data sources to link and analyze safety data to cover at least 100 million U.S. patients. CDER, CBER, and CDRH

developed various approaches to conduct complementary postmarket safety surveillance activities utilizing population-based data sources tailored to the regulated products under the purview of each center. The FDA is consolidating postmarket safety surveillance activities utilizing population-based data sources conducted through the Sentinel 3.0 System, CBER Biologics Effectiveness and Safety (BEST) Initiative, and CDRH under the new Sentinel 3.0. The proposed new system will provide a collaborative, central framework for active surveillance activities for CDER, CBER and CDRH.

The Sentinel 3.0 Data Broker Services Contractor shall serve as the Data Broker, responsible for acquiring, licensing, governing, and providing FDA with access to healthcare data assets required to support the Sentinel 3.0 program and related postmarket safety surveillance activities.

The Contractor shall provide FDA with governed access to commercial healthcare claims data, Healthcare Insurance Claims–EHR linked data, EHR and registry data, Centers for Medicare & Medicaid Services (CMS) data, and outpatient prescription data, consistent with Tiers 1–6 defined in this PWS.

Any access or use granted to external HHS partners shall be subject to applicable data use agreements (DUAs), Memorandum of Understanding (MOU), or FDA governance procedures to ensure protection of data integrity, confidentiality, and compliance with program objectives.

The Sentinel 3.0 program will include the following components:

1. **Data Broker Services (Tiers 1–6):** Provides FDA with governed access to diverse healthcare data sources, including commercial and CMS claims data, outpatient prescription data (subscription with user interface), claims aggregator data, Healthcare Insurance Claims–EHR linked data, specialty EHR data, disease and product registries, registry networks, and Health Information Exchange (HIE) data. Under the Data Broker Services model, the Contractor serves as the single accountable entity responsible for identifying, acquiring, licensing, managing, and governing access to all data assets, including coordinating with third-party data providers as needed. Data sources capable of linkage to Immunization Information Systems (IIS) and medical device utilization data (including Unique Device Identifier (UDI) information) are particularly valuable to FDA operations.
2. **FDA Data Platform:** A Government-developed platform that will provide repository, interoperability, and analytic enablement services for Sentinel 3.0 data. The platform also incorporates AI-enabled data repository services and informatics tools that facilitate data analysis through standardized terminologies and clinical-to-operational translations capabilities.

3. **Studies-As-A-Service:** Provide scientific services including data analytics support for complex pharmacoepidemiology and pharmacovigilance studies, as well as advanced AI-based analytics capabilities.

The FDA Data Platform is a Government-developed, cloud-based data and analytics environment that provides a highly secure, centralized foundation for ingesting, managing, and analyzing data in support of FDA’s public health mission. The platform operates within a Federal Information Security Management Act (FISMA) High Authority to Operate (ATO) environment and is production-ready, scalable, and designed to support enterprise-wide regulatory and analytical needs.

The FDA Data Platform is capable of ingesting and managing the data types described in this PWS, including structured and semi-structured healthcare data such as commercial healthcare insurance claims, claims–EHR linked data, registry data, Centers for Medicare & Medicaid Services (CMS) data, and outpatient prescription data, in accordance with FDA standards for data security, privacy, and interoperability. This description is provided for situational awareness only and does not create additional requirements for the Data Broker Services Contractor beyond those explicitly stated in this PWS.

A single Data Broker Services Contractor will be responsible for acquiring, licensing, governing, and providing FDA access to healthcare data required to support Sentinel 3.0 post-market safety surveillance activities. This includes managing access to data across Tiers 1–6, whether through proprietary data assets, third-party data sources, or a combination thereof, while retaining full contractual, legal, and compliance responsibility for all data delivered under this PWS.

Table 1: Sentinel 3.0 Scope

Tier(s)	Grouped Tier(s)	Topic Area(s)
Tier 1	Data Broker Services	Commercial Healthcare Insurance Claims Data
Tier 2		Commercial Healthcare Insurance Claims Data from Third-Party Aggregator(s) (access to medical records is not required)
Tier 3		Healthcare Insurance Claims-EHR Linked Data
Tier 4		EHR Data (including oncology, pediatrics, geriatrics, inpatient settings of care), Registry or Registry Network Data (product or disease-based registries), and Health Information Exchange (HIE) Data
Tier 5		Centers for Medicare and Medicaid Services (CMS) Data
Tier 6		Outpatient Prescription Data (subscription with user interfaces and raw data)

Scope

Sentinel 3.0 will be led by the FDA Office of the Commissioner. A key element of the Sentinel 3.0 operational framework is the establishment of a flexible and cost-effective data acquisition solution that provides FDA with access to high-quality, fit-for-purpose healthcare data across multiple domains.

Under the Data Broker Services model, the Contractor shall provide FDA with access to all data made available through the FDA Data Platform in support of Sentinel 3.0. The Contractor shall be issued task orders to provide broker-managed data acquisition and access services consistent with the Tier(s) specified in each task order.

The Contractor may subcontract with or otherwise broker access to third-party data providers within the scope of the awarded Tier(s). The Contractor shall retain full contractual, legal, and compliance responsibility for all brokered data assets, including management of data use agreements (DUAs), data governance, privacy protections, and delivery of data to the FDA Data Platform.

The Contractor shall coordinate with FDA and other Sentinel 3.0 support contractors, including the Studies-As-A-Service contractor, as required to support study design, data access, and analytic execution.

The following is a high-level description of key functions of Sentinel 3.0 components to manage a typical request for Sentinel 3.0 services (e.g., scientific consultations, requests for data to address regulatory questions).

- The FDA will receive requests for Sentinel 3.0 services (e.g., scientific questions, requests for information, or consultation) from FDA staff. These requests will be triaged and as appropriate, initiated as new projects and tracked through the Sentinel 3.0 governance process. In collaboration with the Studies-As-A-Service component, the Government will define an overall approach to address each request, including identifying data needs and determining when engagement of the Data Broker Services Contractor is required.
- FDA staff, the Studies-As-A-Service component, and the Data Broker Services Contractor shall collaborate to identify the most appropriate data resources and analytic approaches to address the scientific question. This collaboration includes development of the study protocol and methods, including the Statistical Analysis Plan (SAP) and any required programming or data specifications.
- The Data Broker Services Contractor shall be responsible for acquiring, preparing, and providing access to the required data assets in accordance with the approved

protocol, including performing applicable data quality checks and delivering data or study-specific outputs in a timely manner.

- The FDA Data Platform provides a secure environment through which the Data Broker Services Contractor, the Studies-As-A-Service component, and FDA staff may exchange data and analytic outputs. Data made available through the platform may include broker-managed licensed datasets or study-specific datasets generated in response to approved surveillance or analytic requests.
- The Studies-As-A-Service component is responsible for conducting secondary quality review of analytic outputs, integrating results across datasets when applicable, and supporting interpretation and synthesis of findings for FDA regulatory decision-making.

1 REQUIREMENTS

Independently and not as an agent of the Government, the Contractor shall furnish all the necessary services, qualified personnel, material, equipment, and facilities, not otherwise provided by the Government, as needed to perform the following:

1.1 TASK AREA 1 – DATA BROKER SERVICES

The Government requires primary and continuous Data Broker for the Sentinel 3.0 Program, responsible for acquiring, licensing, governing, and providing FDA access to healthcare data across all applicable Task Areas, including but not limited to Tiers 1–6.

The Contractor may fulfill these requirements through a combination of proprietary data assets and third-party data sources, provided that the Contractor retains full contractual, legal, and compliance responsibility for all data delivered under this PWS.

This task area encompasses the assessment, acquisition, and preparation of diverse healthcare datasets to support the FDA Sentinel 3.0 Program. The Contractor shall be responsible for identifying, evaluating, and procuring commercial claims, Healthcare Insurance Claims-EHR Linked Data (EHR), specialty datasets, registry data, outpatient prescription data, or Centers for Medicare & Medicaid Services (CMS) claims data. It is important that the Contractor have access to medical charts at the patient level for review, abstraction, and adjudication. The Contractor shall ensure that all acquired datasets are compatible with the Sentinel 3.0 FDA data platform and can be linked across sources using privacy-preserving record linkage (PPRL) methods.

The Contractor shall work in close coordination with the Government and other Sentinel 3.0 support contractors (e.g., platform operations and Studies-As-A-Service) to ensure that data assets are delivered in formats and structures consistent with FDA Sentinel 3.0

standards for ingestion, normalization, and integration. All data acquisition activities shall comply with applicable federal laws, regulations, and data use agreements.

1.2 TASK AREA 2 – REQUIRED DATA SERVICES (TIERS 1 – 6)

- 1.1. The Contractor shall establish and operate an enterprise-level Executive Leadership Team to oversee all aspects of Data Acquisition operations.
- 1.2. The Contractor shall establish and maintain business partnerships and DUAs to ascertain electronic healthcare data of tens of millions patient lives.
- 1.3. Under this single-award Data Broker Services model, the Contractor shall be capable of fulfilling requirements across any or all Tiers 1–6 through proprietary assets and/or brokered licensing arrangements.
 - 1.3.1. **Tier 1 - Commercial Claims Data – Direct Sources:** The Contractor shall assess and acquire commercial healthcare insurance claims data covering approximately tens of millions patient lives including both open and closed claims from direct payer sources. The Contractor shall evaluate candidate data sources based on FDA Sentinel 3.0 criteria, negotiate acquisition terms, and ensure datasets are delivered in formats that support ingestion into the FDA Sentinel 3.0 platform. For CBER studies, the Contractor should have DUAs with state/local Immunization Information Systems (IIS) to enhance vaccine exposure information through claims/IIS data linkages.
 - 1.3.2. **Tier 2 - Commercial Claims Data – Aggregators:** The Contractor shall assess and acquire commercial healthcare insurance claims data covering approximately 100 million patient lives from qualified third-party data aggregators. This shall include validation of the aggregator’s sourcing methodology, data coverage, refresh rates, and alignment with Sentinel 3.0 data quality standards.
 - 1.3.3. **Tier 3 - Healthcare Insurance Claims-EHR Linked Data:** The Contractor shall assess and acquire EHR data covering approximately tens of millions of patients’ lives from diverse care settings, including ambulatory, inpatient, and specialty care environments. Candidate datasets shall be evaluated for completeness, representativeness, and compatibility with Sentinel 3.0 data models.
 - 1.3.4. **Tier 4 - EHR, Registry and HIE Data:** The Contractor shall assess and acquire EHR datasets (including oncology, pediatrics, geriatrics, inpatient care), product- or disease-specific registries, registry networks, HIE data as needed to support Sentinel 3.0 scientific and regulatory priorities.
 - 1.3.5. **Tier 5 - CMS Data:** The Contractor shall establish and manage information access (IA) and DUAs with CMS for the acquisition of Medicare and/or Medicaid claims data. The Contractor shall ensure that all CMS-sourced datasets are received, stored, and used in accordance with CMS requirements and applicable

privacy regulations. The Contractor shall ensure that claims data acquired directly from CMS can be linked to commercial data sources within FDA Sentinel 3.0 using PPRL methods. The Contractor shall validate linkage quality and document methodologies for FDA review.

- 1.3.6. Tier 6 – Outpatient Prescription Drug Use Data:** Provide patient-level data describing outpatient prescription drug use (including retail, specialty, and outpatient clinics) in individuals longitudinally. The point-of-sale near real-time pharmacy transaction data shall provide immediate automated access with the ability to provide counts of current and historical aggregate national and sub-national drug utilization information. The longitudinal data shall include information at the prescription claim levels with the ability of characterizing concomitant drug use and characterizing duration of drug exposure. The data shall capture all pharmacy transaction types including cash payments and coupons.
- 1.3.6.1. Provide de-identified, cleaned, ready-to-go, data that can be ingested into the FDA Data platform consisting of nationally based, outpatient prescription-level data that currently exist, commercially available, HIPAA-compliant format. The data shall include historical prescription information at a national and subnational level.
- 1.3.6.2. Provide a description of the Contractor's data acquisition process for all data in all Tiers. All data flow and management procedures shall be described, including data acquisition, cleaning, and editing procedures, data deidentification procedures, specification, and description of all coding systems used for diagnoses, procedures, and drugs, including versions (e.g. ICD-9, ICD-10), methods for coding data, percentage of missing or incomplete information and the method of coding missing data, availability of raw/sample numbers as well as projected numbers, and the frequency with which data are updated and methods for updates. For each health care setting, the Contractor shall estimate the proportion captured in the data source out of the entire market and provide this estimation to FDA, updated at least semi-annually. A current and complete updated data dictionary and database description shall be provided annually and as changes are made.
- 1.3.6.3. The Contractor shall maintain access throughout the period of performance to the applicable data resources (e.g., Commercial Healthcare Insurance Claims, Claims–EHR Linked Data, Registry Data, CMS Data, Outpatient Prescription Data) for the awarded Tier(s) as described by the Contractor in its submitted Data Characteristics Table (Attachment H). The Data Characteristics Table shall serve as a binding representation of the Contractor's proposed data assets for the awarded Tier(s), including but not limited to data provenance, population coverage, refresh frequency, linkage capabilities, historical depth, and applicable limitations. The

Contractor shall notify the Government in advance of any material changes to the data resources described in the Data Characteristics Table and shall obtain Government approval prior to substituting or materially modifying any data asset relied upon to meet Task Area 1 requirements.

- 1.3.6.4. Provide the total number of years of data available overall and the number of years available and accessible in real-time to FDA personnel. A minimum total of ten (10) years of data (minimum of five (5) years of data available online and at a minimum an additional five (5) years of historical data that is no longer online) and propose methods to obtain historical data in a timely manner. FDA's preference is that historical datasets be available on demand and without intervention from the Contractor.
- 1.3.6.5. Provide prescription-level data with the ability of analyzing current and historical aggregate national and sub-national dispensed prescription information at a minimum of monthly level, with the ability to aggregate data by time periods specified by the user. Patient level data shall provide the ability to analyze unique patients utilizing drug products in an outpatient setting, with the capability of stratifying by patient age and sex. Data shall have the capability of customizing patient age bands as needed. Patient-level data shall link to prescription-level data, allowing for the assessment of new versus continuing use of drugs at the molecule level and at the drug class level.
- 1.3.6.6. Provide the size and characteristics of the population of patients or covered lives from which the prescription-and patient-level data are derived, since these can affect the quality and usefulness of the data provided and the validity of national projections from the data. Characteristics shall minimally include: number of individual patients or covered lives per calendar quarter; number and percent of patients or covered lives with at least two years of continuous activity; number and percent of patients or covered lives with corresponding enrollment information; availability of enrollment/eligibility/participant inclusion information for use in data analyses; number and percentage of active patients or covered lives in populations of special interest to FDA (including children [ages ≤ 17 years], females of child-bearing age, and elderly [ages ≥ 65 years]); turnover or churn rate for participation in the data resource; the method by which patient age is obtained; geographic diversity; patient demographics; coverage or inclusion policies relevant to pharmaceutical products under health care plans providing the majority of the data; and availability of medical claims data. The number and percent of the total population for whom clinical/medical data or data capturing the administration of drug or biologic products in the outpatient setting by healthcare professionals are available and detailed information describing how this information is linked to the

pharmacy data is required. These longitudinal data shall follow uniquely identifiable individuals (using encrypted patient identifiers) and healthcare providers over time either across or within ambulatory health care plans. Patient age and sex shall be provided. Information on race/ethnicity shall be accompanied by documentation of the origin of this information, if available.

- 1.3.6.7. Provide prescription data with the therapeutic class of drug and biologic products (including biosimilars and interchangeable biosimilars), molecule(s), active ingredient(s) (e.g., single-ingredient or combination), molecular salt (if applicable), product name (U.S. proprietary and non-proprietary), original prescribed product, substitution allowability, strength, National Drug Code (NDC), company name (manufacturer or corporation), dosage form, strength, all dates dispensed, quantity dispensed, dispensed days' supply, prescriber specialty, and method of payment. The Contractor shall provide the ability to obtain all data elements within a single dataset extract. The Contractor shall provide a history of dataset adjustments and corrections over the past 3 years.
- 1.3.6.8. Capability to provide medical device identification, including unique device identification (UDI) at the patient-level and device-level data and to append data provided historically and future with medical device identification, including UDI data.
- 1.3.6.9. Provide direct access to weekly data with the capability of providing current and historical national-level and sub-national level counts of prescriptions and unique patients over time, delivered weekly with no more than an eight calendar (8) day lag.
- 1.3.6.10. Provide direct access to monthly data with the capability of providing current and historical national-level and sub-national level counts of prescriptions and unique patients over time, delivered monthly with no more than a thirty (30) calendar day lag.
- 1.3.6.11. Capability to provide data in weekly, monthly, quarterly, and annual outputs.
- 1.3.6.12. Ability to use both the prescription-based outpatient data and patient-based outpatient data to estimate drug utilization patterns in the total U.S. outpatient population, or in a national sub-population of interest (e.g., children [≤ 17 years], females of child-bearing age, elderly [≥ 65 years]). This shall require the use of projection or other appropriate methodologies which shall be valid, reliable, and robust regarding drug utilization specifically. The methods of national level estimation (projection or other methodology) shall be fully disclosed.
- 1.3.6.13. Provide claims-level data, such as prescriptions or medical claims including a de-identified patient ID for linking patients across claims and a de-identified health care practitioner ID for linking practitioners across claims. The FDA has no

interest in knowing the identity of any patient, provider, or health plan included in the data. It shall be solely the Contractor's responsibility to de-identify the data for FDA use, but the methods used to de-identify the data shall be described.

- 1.3.6.14. Provide mother-infant linked records. In order to evaluate the effect of exposures in pregnancy on outcomes in infants, it is necessary to link mothers/deliveries with infants.
- 1.3.6.15. Provide access through an existing interface and this interface shall be user-friendly, transparent, and flexible enough to meet the FDA's needs. Minimally, the interface shall provide both predefined (e.g., canned) and custom reports.
- 1.3.6.16. Provide the FDA (or work with the FDA data platform to confirm) the ability to immediately, directly download data in an automated fashion—without any intervention needed from the Contractor in the following formats: Microsoft Excel, SAS, R, comma-or tab-separated file, and text delimited file.
- 1.3.6.17. The Contractor may need to provide additional data analyses, custom queries, or raw data preparation not available through the data tools and in increments to be determined at the Task Order level.

1.4. Prior to acquisition, the Contractor shall evaluate all candidate datasets using standardized FDA Sentinel 3.0 criteria, including but not limited to:

- Uniqueness of individuals
- Data type and structure
- Unique device identification (UDI)
- Clinical coding systems used
- Diagnosis and procedure coverage
- Refresh rate and historical depth
- Population representativeness
- Linkage potential
- Price and licensing terms

The Contractor shall provide FDA with documented evaluation results and recommendations for acquisition.

- 1.5. The Contractor shall develop templated, reproducible DUAs that allow permissible use across Sentinel 3.0 program activities. These DUAs shall include standardized terms to streamline FDA tracking, renewals, and compliance monitoring.
- 1.6. The Contractor shall gather the data - defining a clear objective, identity reliable data sources, ensure data accuracy and privacy, and establish a method for collection. The Contractor shall harmonize the data aligning and standardizing data from multiple sources to ensure accuracy, and usability across the business usages.

1.7. The Contractor shall develop and maintain a data onboarding roadmap that documents dataset prioritization schemes and supports FDA decision-making for data acquisition, providing transparency to stakeholders on the process for making datasets available. The roadmap must include an on-demand dashboard accessible to designated FDA personnel, displaying current dataset status in the ingestion process, projected availability dates, timelines, estimated scope and level of effort, notable roadblocks, and onboarding costs. The Contractor shall provide the capability to subset or assemble analytic cohorts across multiple data sources. All cohort definitions shall include documented provenance, reproducible query parameters, and metadata sufficient for regulatory and scientific transparency.

1.3 TASK AREA 3 – ADDITIONAL DATASET ACQUISITION AND ACCESS SERVICES (OPTIONAL)

This Task Area is optional and may be exercised at the sole discretion of the Government. Under this Task Area, the Contractor shall, when authorized by the Government through a Task Order, identify, acquire, license, and provide access to additional healthcare datasets or data access services that augment or complement the Tier 1–6 datasets defined under Task Area 2.

These services are intended to provide the Government with flexibility to obtain supplemental, emerging, or specialized data assets needed to address evolving regulatory, scientific, or public health questions that are not fully supported by the core Tier 1–6 data holdings.

This Task Area does not replace or modify the Contractor’s responsibilities for delivering Tier 1–6 data under the Data Broker model. All activities under this Task Area shall be explicitly scoped, approved, and governed through Government-issued Task Orders.

The Government may authorize optional data acquisitions under this Task Area by issuing Task Orders at its discretion. The Government may issue up to five (5) separate Task Orders per year not to exceed \$25,000,000 over the period of performance of this IDIQ. Each exercised option increment shall specify, at a minimum, the supplemental dataset(s) or data access service(s) authorized, the scope and intended use of the data, and the applicable period of performance.

The Government is not obligated to exercise this Task Area or any associated optional CLINs. No work under this Task Area shall commence unless and until the Government exercises the applicable option increment in writing and provides funding authorization.

The Contractor shall be capable of:

1.8. Identifying and evaluating additional data assets or data access services, including but not limited to:

- Supplemental commercial datasets
- Specialty or niche datasets
- Registry or network-based datasets
- Time-sensitive or emerging data sources

1.9. Procuring, licensing, and managing access to approved supplemental datasets in accordance with applicable data use agreements, privacy protections, and federal requirements.

1.10. Providing access to such datasets through appropriate delivery or access mechanisms, which may include:

- Direct data delivery for ingestion into the FDA Data Platform
- API-based access
- Federated or query-based access models

The specific access method shall be determined by the Government at the Task Order level based on the use case, data characteristics, and governance requirements.

1.11. Supporting use cases such as:

- Augmenting Tier 1–6 datasets for targeted analyses
- Rapid-response or time-sensitive regulatory inquiries
- Exploratory analyses to assess feasibility or signal detection
- Pilot or proof-of-concept evaluations of new data sources

All data acquired or accessed under this Task Area shall comply with FDA Sentinel 3.0 data standards, security and privacy requirements, and applicable governance controls.

2 CONTRACT CONSTRAINTS

The following constraints are applicable to this contract:

1. Organizational Conflict of Interest (OCI)

An organizational conflict of interest may arise when the Contractor's relationships with data providers, subcontractors, or affiliates could bias performance under this contract (e.g., steering FDA toward certain datasets, limiting access to alternatives, or using nonpublic information for competitive advantage). The Contractor shall

identify, disclose, and mitigate potential OCIs in accordance with FAR Subpart 9.5 and the requirements of this contract.

FDA's mitigation strategy to avoid potential organizational conflicts of interest shall require all Offerors submitting proposals, as prime Contractors or as a sub-Contractor must conduct an in-depth assessment for the potential of an organizational conflicts of interest.

2. Advance Understanding

Notwithstanding any of the above provisions, under certain circumstances, the Government may be required to share data or information it obtains under this contract with Congress, with a federal court, or with another governmental body of competent jurisdiction. If the Government intends to share data or information obtained under this contract with Congress, with a federal court, or with another governmental body of competent jurisdiction, the Government will take appropriate measures within its control to ensure that the information will be shared in a manner intended to protect the information from public disclosure contract shall not constrain the FDA from releasing study results for regulatory issues of public health importance prior to formal publication or presentation in a scientific forum. Under these circumstances, FDA will, to the extent practicable, work with the Contractor to avoid jeopardizing timely publication of results.

3. Confidential or Otherwise Privileged Information

The Contractor shall assure the protection of confidential or otherwise privileged information disclosed by the FDA from unauthorized use or disclosure and must treat the information as confidential or otherwise privileged.

Confidential or otherwise privileged information disclosed shall not be used by the Contractor to solicit or in any other way encourage pharmaceutical or medical device industry funding/financial sponsorship of research projects to be performed by the Contractor. In the event that an outside party such as a drug or biologics NDA or BLA sponsor, or a medical device PMA, 510k or other sponsor, independently requests that the Contractor conduct research on a safety issue that FDA has previously discussed with the Contractor but FDA has not requested the Contractor perform the research on that issue, the Contractor may conduct such research only if both of the following conditions are satisfied: (1) the Contractor shall not divulge confidential and privileged information previously disclosed by the FDA; and (2) the Contractor shall recuse itself from all subsequent conversations with FDA about the safety of the involved drug, medical device or biologic product(s). In conjunction with the above, the Contractor shall provide a Conflict-of-Interest Plan as outlined in FDA Clause 1337 Conflict of interest (see FDA 1337, Conflict of Interest).

4. Clearance for Publication

No information furnished to or generated by the Contractor in the performance of this contract shall be released to the public until it has been reviewed by the FDA COR for accuracy of factual data and interpretation. The FDA shall have 30 calendar days after receipt of the manuscript or other written draft of the material to be released, to review, comment, and return to the Contractor.

5. IT Compliance Requirements

The Contractor shall be responsible for knowledge of and compliance with all applicable federal information technology and information management laws, regulations, policies, and standards at the government-wide, HHS and FDA levels. At the government-wide level, these include Office of Management and Budget (OMB), National Institute of Standards and Technology (NIST), and General Accounting Office (GAO). These can be primarily found at or through the Federal CIO Council website at: <http://www.cio.gov/>. HHS documents are found at: <http://www.hhs.gov/oirm/>

The computer programs, data management systems, databases, and other resources used or developed through this contract shall comply with data standards and procedures established by DHHS and FDA (per <http://www.fda.gov/ForIndustry/DataStandards/default.htm>) .

6. Section 508 Compliance

The Contractor shall be familiar with Section 508 requirements as described at <http://www.section508.gov/> in order to ensure that the deliverables generated as part of this contract shall be compliant with Section 508. The complete text of Section 508 Final Provisions can be accessed at <https://www.access-board.gov/ict/> and <https://www.hhs.gov/web/governance/digital-strategy/it-policy-archive/hhs-policy-section-508-compliance-accessibility-information-communications-technology.html>.

3 GOVERNMENT-FURNISHED PROPERTY, EQUIPMENT, INFORMATION (GFP/GFE/GFI)

The Government furnished property (GFP) (materials, equipment, and/or information) may be provided in conjunction with required performance under individual task orders as determined by mutual agreement between FDA and the Contractor.

For some Sentinel 3.0 Contractors, the FDA may provide Government-Furnished Property (GFP)/Government-Furnished Equipment ((GFE)/Government-Furnished Information (GFI) such as laptops and the associated PIV card badges, security badges, computer monitors,

SOPs/Work Guides, for the performance of work under this contract/task order. Laptops shall not be used for any other work outside this contract/task order.

The Contractor shall be liable for loss or damage to GFP/GFE beyond fair wear and tear in accordance with the Government Furnished Property clause of the contract.

The Contractor shall maintain an up-to-date full inventory list of GFP/GFE/GFI in the Contractor Roster in accordance with government regulations on a monthly basis to the COR/Program Manager (PM) via an excel spreadsheet in SharePoint. At a minimum, the list shall include all GFP/GFE/GFI, the Contractor employee's name, email address, and labor category position, property number.

4 SECURITY AND PRIVACY

Security and Privacy Requirements

1. Baseline Security Requirements

a. **Applicability.** The requirements herein apply whether the entire contract or modification (hereafter "contract"), or portion thereof, includes either or both of the following:

i. **Access (Physical or Logical) to Government Information:** A Contractor (and/or any subcontractor) will have or will be given the ability to have, routine physical (entry) or logical (electronic) access to government information.

ii. **Operate a Federal System Containing Information:** A Contractor (and/or any subcontractor) will operate a federal system and information technology containing data that supports the FDA mission. In addition to the Federal Acquisition Regulation (FAR) Subpart 2.1 definition of "information technology" (IT), the term as used in this section includes computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the central processing unit of a computer, software, firmware and similar procedures, services (including support services), and related resources.

b. **Safeguarding Information and Information Systems.** All government information and information systems must be protected in accordance with FDA policies and level of risk. At a minimum, the Contractor (and/or any subcontractor) must:

i. Protect the:

Confidentiality, which means preserving authorized restrictions on access and disclosure, based on the security terms found in this contract, including means for protecting personal privacy and proprietary information.

Integrity, which means guarding against improper information modification or destruction, and ensuring information non-repudiation and authenticity; and

Availability, which means ensuring timely and reliable access to and use of information. **Note to the Requiring Activity Representative:** Complete the following section using the information obtained from the Information Security and Privacy Certification Checklist.

ii. Categorize all information owned and/or collected/managed on behalf of FDA and information systems that store, process, and/or transmit FDA information in accordance with FIPS 199 and National Institute of Standards and Technology (NIST) Special Publication (SP) 800-60, Volume II: *Appendices to Guide for Mapping Types of Information and Information Systems to Security Categories*. Based on information provided by the System/Data Owner, ISSO, privacy representative, or other POC, the impact level for each Security Objective (Confidentiality, Integrity, and Availability) and the Overall Impact Level, which is the highest watermark of the three factors of the information or information system are the following:

Confidentiality: ☐ Low ☒ Moderate ☐ High

Integrity: ☐ Low ☒ Moderate ☐ High

Availability: ☐ Low ☒ Moderate ☐ High

Overall Impact Level: ☐ Low ☒ Moderate ☐ High

iii. Based on the agreed-upon level of impact, implement the necessary safeguards to protect all information systems and information collected and/or managed on behalf of FDA regardless of location or purpose.

iv. Report any discovered or unanticipated threats or hazards by either the agency or Contractor, or if existing safeguards have ceased to function immediately after discovery, **within one (1) hour or less**, to the government representative(s). This includes notifying the FDA Cybersecurity and Infrastructure Operations Coordination Center (CIOCC) within one (1) hour of discovery/detection in the event of a cybersecurity or privacy incident.

v. Adopt and implement all applicable policies, procedures, controls, and standards required by the FDA Information Security Program to ensure the confidentiality, integrity, and availability of government information and government information systems for which the Contractor is responsible under this contract or to which the Contractor may otherwise have access under this contract. Obtain the FDA Information Security Program security

requirements, outlined in the FDA Information Security and Privacy protection (IS2P) policy, by contacting the CO/COR or emailing your ISSO. etc. **Privacy Act.** Comply with the Privacy Act requirements (when applicable), and tailor FAR and HHSAR clauses as needed. ed. **Privacy Compliance.** Comply with the E-Government Act of 2002, NIST SP 800-53, and applicable FDA privacy policies and complete all the requirements below:

Note to the Requiring Activity Representative: Complete this section using the information obtained from the Information Security and Privacy Certification Checklist. This information may be included after award in the event it is not yet available at the time of acquisition.

i. Per the Office of Management and Budget (OMB) Circular A-130, Personally Identifiable Information (PII), is "information that can be used to distinguish or trace an individual's identity, either alone or when combined with other information that is linked or linkable to a specific individual." Examples of PII include, but are not limited to the following: Social Security number, date and place of birth, mother's maiden name, biometric records, etc.

ii. Based on information provided by the ISSO, System/Data Owner, or other security or privacy representative, it has been determined that this solicitation/contract involves:

☐ No PII ☒ PII

iii. The Contractor must support the agency with conducting a Privacy Threshold Analysis (PTA) for the information system and/or information handled under this contract to determine whether or not a full Privacy Impact Assessment (PIA) needs to be completed.

If the results of the PTA show that a full PIA is needed, the Contractor must support the agency with completing a PIA for the system or information after completion of the PTA and in accordance with HHS and FDA policy and OMB M-03-22, *Guidance for Implementing the Privacy Provisions of the E-Government Act of 2002*. The PTA/PIA must be completed and approved prior to active use and/or collection or processing of PII and is a prerequisite to agency issuance of an authorization to operate (ATO).

The Contractor must support the agency in reviewing the PIA at least every **three years** throughout the system development lifecycle (SDLC)/information lifecycle, or when determined by the agency that a review is required based on a major change to the system, or when new types of PII are collected that introduces new or increased privacy risks, whichever comes first.

e. **Controlled Unclassified Information (CUI).** Executive Order 13556 defines CUI as "information that laws, regulations, or Government-wide policies require to have

safeguarding or dissemination controls, excluding classified information." The Contractor (and/or any subcontractor) must comply with *Executive Order 13556, Controlled Unclassified Information, (implemented at 3 CFR, part 2002)* when handling CUI. 32 C.F.R. 2002.4(aa) As implemented the term "*handling*" refers to "...any use of CUI, including but not limited to marking, safeguarding, transporting, disseminating, re-using, and disposing of the information." 81 Fed. Reg. 63323. The requirements below apply only to nonfederal systems that process, store, or transmit CUI, or that provide security protection for such components. All sensitive information that has been identified as CUI by a regulation or statute, handled by this solicitation/contract, must be:

- i. Marked appropriately.
- ii. Disclosed to authorized personnel on a Need-To-Know basis.
- iii. Protected in accordance with NIST SP 800-53, *Security and Privacy Controls for Information Systems and Organizations* applicable baseline if handled by a contractor system operated on behalf of the agency, or NIST SP 800-171, *Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations* if handled by internal Contractor system; and
- iv. Returned to FDA control, destroyed when no longer needed, or held until otherwise directed. Information and/or data must be disposed of in accordance with NIST SP 800-88, *Guidelines for Media Sanitization*.

f. Protection of Sensitive Information. For security purposes, information is or may be sensitive because it requires security to protect its confidentiality, integrity, and/or availability. The Contractor (and/or any subcontractor) must protect all government information that is or may be sensitive by securing it with a solution that is validated with current FIPS 140 validation certificates from the NIST CMVP.

g. Government Furnished Equipment (GFE) for Foreign Travel. FDA personnel are prohibited from taking GFE when participating in personal, unofficial travel to foreign countries. FDA personnel are strictly prohibited from teleworking using GFE in foreign countries. FDA personnel must also request loaner GFE from the FDA Foreign Travel

h. Confidentiality and Nondisclosure of Information. Any information provided to the Contractor (and/or any subcontractor) by FDA or collected by the Contractor on behalf of FDA must be used only for the purpose of carrying out the provisions of this contract and must not be disclosed or made known in any manner to any persons except as may be necessary in the performance of the contract. The Contractor assumes responsibility for protection of the confidentiality of Government records and must ensure that all work performed by its employees and subcontractors must be under the supervision of the

Contractor. Each Contractor employee or any of its subcontractors to whom any FDA records may be made available or disclosed must be notified in writing by the Contractor that information disclosed to such employee or subcontractor can be used only for that purpose and to the extent authorized herein. The confidentiality, integrity, and availability of such information must be protected in accordance with HHS and FDA policies. Will be subject to the HHS and FDA sanction policies and/or governed by the following laws and regulations: ii. 18 U.S.C. 641 (Criminal Code: Public Money, Property or Records); iii. 18 U.S.C. 1905 (Criminal Code: Disclosure of Confidential Information); and

iii. 44 U.S.C. Chapter 35, Subchapter I (Paperwork Reduction Act).

i. **Internet Protocol Version 6 (IPv6).** All procurements using Internet Protocol must comply with OMB Memorandum M-05-22, *Transition Planning for Internet Protocol Version 6 (IPv6)*.

j. **Information and Communications Technology (ICT).** ICT products and services from prohibited entities/sources must not be used/acquired in compliance with Public Law 115-232, Section 889 Parts A and B, FAR 4.21, FAR 52.204.23, FAR 52.204.24, and FAR 52.204.25. The Contractor (and/or any subcontractor) must notify the government if they identify prohibited ICT products and/or services are used during the contract performance.

k. **Government Websites.** All new and existing public-facing government websites must be securely configured with Hypertext Transfer Protocol Secure (HTTPS) using the most recent version of Transport Layer Security (TLS). In addition, HTTPS must enable HTTP Strict Transport Security (HSTS) to instruct compliant browsers to assume HTTPS at all times to reduce the number of insecure redirects and protect against attacks that attempt to downgrade connections to plain HTTP. For internal-facing websites, HTTPS is not required, but it is highly recommended. Consult the *HHS Policy for Internet and Email Security* for additional information.

l. **Contract Documentation.** The Contractor must use provided templates, policies, forms, and other agency documents to comply with contract deliverables as appropriate.

Note to the Requiring Activity Representative: See Appendix C for baseline deliverables. Do **NOT** include in procurement documentation.

m. **Standard for Encryption.** The Contractor (and/or any subcontractor) must:

i. Comply with the *HHS Standard for Encryption of Computing Devices and Information* to prevent unauthorized access to government information.

ii. Encrypt all sensitive federal data and information (i.e., PII, protected health information [PHI], proprietary information, etc.) in transit (i.e., email, network connections, etc.) and at

rest (i.e., servers, storage devices, mobile devices, backup media, etc.) with an encryption solution that is validated with current FIPS 140 validation certificates from the NIST CMVP.

iii. Secure all devices (i.e.: desktops, laptops, mobile devices, etc.) that store and process government information and ensure devices meet HHS and FDA-specific encryption standard requirements. Maintain a complete and current inventory of all laptop computers, desktop computers, and other mobile devices and portable media that store or process sensitive government information (including PII).

iv. Verify that the encryption solutions in use have been validated under the Cryptographic Module Validation Program to confirm compliance with current FIPS 140 validation certificates from the NIST CMVP. The Contractor must provide a written copy of the validation documentation to the COR.

v. Use the Key Management system on the HHS personal identification verification (PIV) card or establish and use a key recovery mechanism to ensure the ability for authorized personnel to encrypt/decrypt information and recover encryption keys <http://csrc.nist.gov/publications/>. Encryption keys must be provided to the COR upon request and at the conclusion of the contract.

n. **Contractor Non-Disclosure Agreement (NDA).** Each Contractor (and/or any subcontractor) employee having access to non-public government information under this contract must complete the FDA non-disclosure agreement (3398 Form)], as applicable. Contractors (and/or subcontractors) must submit a copy of each signed and witnessed NDA to the Contracting Officer (CO) and/or CO Representative (COR) prior to performing any work under this acquisition.

Note to the Requiring Activity Representative: See Appendix D for the FDA Contractor Non-Disclosure Agreement. Do **NOT** include in procurement documentation.

2. Training Requirements

a. **Mandatory Training for All Contractor Staff.** All Contractor (and/or any subcontractor) employees assigned to work on this contract must complete the applicable FDA information security awareness, privacy, and records management training (provided upon contract award) before performing any work under this contract. Thereafter, the employees must complete FDA information security awareness, privacy, and records management training at least **annually**, during the life of this contract. All provided training must be compliant with HHS training policies.

b. **Role-based Training.** All Contractor (and/or any subcontractor) employees with significant security responsibilities (as determined by the program manager) must

complete role-based training **annually** commensurate with their role and responsibilities in accordance with HHS and FDA policy.

c. **Training Records.** The Contractor (and/or any subcontractor) must maintain training records for all its employees working under this contract in accordance with HHS and FDA policy. A copy of the training records must be provided to the CO and/or COR within **30 days** after contract award and **annually** thereafter or upon request.

3. Rules of Behavior

a. The Contractor (and/or any subcontractor) must ensure that all employees performing on the contract comply with the *HHS Information Technology General Rules of Behavior*, *HHS Rules of Behavior for Privileged Users*, and FDA policies and standards.

b. All Contractor employees performing on the contract must read and adhere to the Rules of Behavior before accessing Agency data or other information, systems, and/or networks that store/process government information, initially at the beginning of the contract and at least **annually** thereafter, which may be done as part of annual FDA Information Security Awareness Training. If the training is provided by the Contractor, the signed ROB must be provided as a separate deliverable to the CO and/or COR per defined timelines above.

4. Incident Response

- a. The Contractor (and/or any subcontractor) must respond to all alerts/Indicators of Compromise (IOCs) provided by HHS Computer Security Incident Response Center (CSIRC)/FDA CIOCC /Incident Response Team teams **within 24 hours**, whether the response is positive or negative.

FISMA defines an incident as "an occurrence that (1) actually or imminently jeopardizes, without lawful authority, the integrity, confidentiality, or availability of information or an information system; or (2) constitutes a violation or imminent threat of violation of law, security policies, security procedures, or acceptable use policies. In accordance with OMB M-17-12, *Preparing for and Responding to a Breach of Personally Identifiable Information (PII)*, an incident is "an occurrence that (1) actually or imminently jeopardizes, without lawful authority, the integrity, confidentiality, or availability of information or an information system; or (2) constitutes a violation or imminent threat of violation of law, security policies, security procedures, or acceptable use policies" and a privacy breach is "the loss of control, compromise, unauthorized disclosure, unauthorized acquisition, or any similar occurrence where (1) a person other than an authorized user accesses or potentially accesses personally identifiable information or (2) an authorized user accesses or potentially accesses personally identifiable information for an other than authorized purpose." For additional information on the HHS breach response process, please see the

FDA IS2P Appendix F: Incident Response and the *HHS Policy and Plan for Preparing for and Responding to a Breach of Personally Identifiable Information (PII)*."

b. In the event of a suspected or confirmed incident or breach, the Contractor (and/or any subcontractor) must:

i. Protect all sensitive information, including any PII created, stored, or transmitted in the performance of this contract, with encryption solution that is validated with current FIPS 140 validation certificates from the NIST CMVP.

ii. NOT notify affected individuals unless so instructed by the Contracting Officer or designated representative. If so, instructed by the Contracting Officer or representative, the Contractor must send FDA approved notifications to affected individuals as directed by FDA's SOP.

iii. Report all suspected and confirmed information security and privacy incidents and breaches to the FDA CIOCC, COR, CO, FDA SOP (or his or her designee), and other stakeholders, including breaches involving PII, in any medium or form, including paper, oral, or electronic, as soon as possible and without unreasonable delay, no later than **one (1) hour**, and consistent with the applicable FDA and HHS policy and procedures, NIST standards and guidelines, as well as US-CERT notification guidelines. The types of information required in an incident report must include at a minimum: company and point of contact information, contact information, impact classifications/threat vector, and the type of information compromised. In addition, the Contractor must:

☐ Cooperate and exchange any information, as determined by the Agency, necessary to effectively manage or mitigate a suspected or confirmed breach.

Not include any sensitive information in the subject or body of any reporting e-mail; and

Encrypt sensitive information in attachments to email, media, etc.

iv. Comply with OMB M-17-12, *Preparing for and Responding to a Breach of Personally Identifiable Information*, and HHS and FDA breach response policies when handling PII breaches.

v. Provide full access and cooperate on all activities as determined by the Government to ensure an effective incident response, including providing all requested images, log files, and event information to facilitate rapid resolution of sensitive information incidents. This may involve disconnecting the system processing, storing, or transmitting the sensitive information from the Internet or other networks or applying additional security controls. This may also involve physical access to Contractor facilities during a breach/incident investigation on demand. i5. Position Sensitivity Designations. All Contractor (and/or any

subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR). The following position sensitivity designation levels apply to this solicitation/contract (e.g. tier 1, 2, or 4): 2.

Note to the Requiring Activity Representative: *The Requiring Activity Representative, in conjunction with Personnel Security, must use the OPM Position Sensitivity Designation automated tool (<https://www.opm.gov/investigations/>) to determine the sensitivity designation for background investigations. After making those determinations, include all applicable position sensitivity designations.*

6. Homeland Security Presidential Directive (HSPD)-12

The Contractor (and/or any subcontractor) and its employees must comply with Homeland Security Presidential Directive (HSPD)-12, *Policy for a Common Identification Standard for Federal Employees and Contractors*; OMB M-05-24; OMB M-19-17; FIPS 201, *Personal Identity Verification (PIV) of Federal Employees and Contractors*; HHS HSPD-12 policy; and Executive Order 13467, Part 1 §1.2.

Note to the Requiring Activity Representative: *For additional information, see HSPD-12 policy at: <https://www.dhs.gov/homeland-security-presidential-directive-12>*

7. Roster

The Contractor (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, have the ability to access, or host and/or maintain a government information system(s). The roster must be submitted to the COR and/or CO per the COR or CO's direction. Any revisions to the roster as a result of staffing changes must be submitted within a timeline as directed by the COR and/or CO. e COR will notify the Contractor of the appropriate level of investigation required for each staff member. If the employee is filling a new position; the Contractor must provide a position description, and the Government will determine the appropriate suitability level. e8. Contract Initiation and Expiration

a. **General Security Requirements.** The Contractor (and/or any subcontractor) must comply with information security and privacy requirements, Enterprise Performance Life Cycle (EPLC) processes, HHS Enterprise Architecture requirements to ensure information is appropriately protected from initiation to expiration of the contract. All information systems development or enhancement tasks supported by the Contractor must follow the FDA EPLC framework and methodology in accordance with the FDA EPLC Project documentation, located here:

<http://epoint.fda.gov/orgs/DelMgmtSupport/IntakeProc/EPLCv2/SitePages/v2/EPLCHome.aspx> and in accordance with the HHS Contract Closeout Guide (2012). eb. **System**

Documentation. Contractors (and/or any subcontractors) must follow and adhere to HHS System Development Life Cycle requirements, at a minimum, for system development and provide system documentation at designated intervals (specifically, at the expiration of the contract) within the EPLC that require artifact review and approval. ec. **Sanitization of Government Files and Information.** As part of contract closeout and at expiration of the contract, the Contractor (and/or any subcontractor) must provide all required documentation in accordance with SMGs published by FDA's Office of Acquisitions and Grant Services (OAGS) to the CO and/or COR to certify that, at the government's direction, all electronic and paper records are appropriately disposed of and all devices and media are sanitized in accordance with NIST SP 800-88, *Guidelines for Media Sanitization*.

d. **Notification.** The Contractor (and/or any subcontractor) must notify the CO and/or COR and system ISSO as soon as it is known that a contract employee will stop working under this contract.

e. **Contractor Responsibilities upon Physical Completion of the Contract.** The Contractor (and/or any subcontractors) must return all government information and IT resources (i.e., government information in non-government-owned systems, media, and backup systems) acquired during the term of this contract to the CO and/or COR. Additionally, the Contractor must provide a certification that all government information has been properly sanitized and purged from Contractor-owned systems, including backup systems and media used during contract performance, in accordance with HHS and Ff. The Contractor (and/or any subcontractor) must perform and document the actions identified in the FDA eDepart system

<http://inside.fda.gov:9003/EmployeeResources/NewEmployee/eDepartDepartureSystem/default.htm> as soon as it is known that a contract an employee will terminate work under this contract. The Contractor (and/or any subcontractor) shall coordinate with the COR via email, copying the Contract Specialist, to ensure that the appropriate person performs and documents the actions identified in the FDA eDepart system. F9. Records Management and Retention Fa. The Contractor (and/or any subcontractor) must maintain all information in accordance with Executive Order 13556 -- Controlled Unclassified Information, National Archives and Records Administration (NARA) records retention policies and schedules and *HHS Policy for Records Management* and HHS and FDA policies and must not dispose of any records unless authorized by HHSFDA.

b. In the event that a Contractor (and/or any subcontractor) accidentally disposes of or destroys a record without proper authorization, he/she must document and report the incident in accordance with HHS and FDA policies.

10. High Value Asset (HVA)

If a system is identified as HVA,²⁴ the Contractor must comply with the FDA IS2P Appendix AB: High Value Asset (HVA) Program, the HHS Policy for the High Value Asset (HVA) Program, and the DHS HVA Control Overlay²⁵ in addition to the above requirements.

All documentation must be available to the CO and/or COR upon request.

1. Security Requirements for GOCO or COCO
2. Resources

a. **Federal Policies.** The Contractor (and/or any subcontractor) must comply with applicable federal laws and HHS and FDA policies that include, but are not limited to, the *HHS Information Security and Privacy Protection (IS2P) policy*; *FDA Information Security and Privacy Protection (IS2P) policy*; *Federal Information Security Modernization Act (FISMA) of 2014, (44 U.S.C. 101)*; National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53, latest revision, *Security and Privacy Controls for Information Systems and Organizations*; Office of Management and Budget (OMB) Circular A-130, *Managing Information as a Strategic Resource*; and other applicable federal laws, regulations, NIST guidance, Departmental, and Agency policies.

- b. **Assessment and Authorization (A&A).** A valid authority to operate (ATO) certifies that the Contractor's information system meets the contract's requirements to protect the agency data. If the system under this contract does not have a valid ATO, the Contractor (and/or any subcontractor) must work with the agency and supply the deliverables required to complete the ATO within the specified timeline(s): *[Within 30 days]*. The Contractor must conduct the A&A requirements in accordance with *HHS IS2P/ FDA IS2P NIST SP 800-37, Guide for Applying the Risk Management Framework to Information Systems: A Security Life Cycle Approach* (latest revision), NIST SP 800-53B, *Control Baselines for Information Systems and Organizations*, and the NIST SP 800-53A (latest revision).

Note to the Requiring Activity Representative: For an existing ATO, the FDA must make a determination if the system needs to be re-authorized to ensure all necessary safeguards are in place to protect the system and information for the performance of the contract and state as such.

FDA acceptance of the ATO does not alleviate the Contractor's responsibility to ensure the system security and privacy controls are implemented and operating effectively.

- i. **A&A Package Deliverables** - The Contractor (and/or any subcontractor) must provide an A&A package within a timeline directed by the COR, per the FDA Enterprise Performance Lifecycle (EPLC) process, to the CO and/or COR. The following A&A deliverables are required to complete the A&A package:
- ii. **System Security Plan (SSP)** - due a week prior to the start of the annual security assessment. The SSP must comply with the NIST SP 800-18, *Guide for Developing Security Plans for Federal Information Systems*, the Federal Information Processing Standard (FIPS) 200, *Recommended Security Controls for Information Systems*, and NIST SP 800-53, *Security and Privacy Controls for Information Systems and Organizations* applicable baseline requirements, and other applicable NIST guidance as well as HHS and FDA policies and other guidance. The SSP must be consistent with and detail the approach to IT security contained in the Contractor's bid or proposal that resulted in the award of this contract. The SSP must provide an overview of the system environment and security requirements to protect the information system as well as describe all applicable security controls in place or planned for meeting those requirements. It should provide a structured process for planning adequate, cost-effective security protection for a system. The Contractor must review and update the SSP at least **annually** thereafter and if requested, provide a copy of the updated SSP.
- iii. **Security Assessment Plan/Report (SAP/SAR)** - due before the system is made available to standard users. The security assessment must be conducted by FDA's team of security assessors, unless otherwise noted and be consistent with NIST SP 800-53A, NIST SP 800-30, and HHS and FDA policies. The assessor will document the assessment results in the

SAR. **Note to the Requiring Activity Representative:** *The FDA should determine which security control baseline applies and then make a determination on the appropriateness/necessity of obtaining an independent assessment. Assessments of controls can be performed by Contractor, government, or third parties, with third-party verification considered the strongest. If independent assessment is required, include statement below.* Thereafter, the Contractor, in coordination with FDA, must assist in the assessment of the security controls and update the SAR at least **annually**. A copy of the updated SAR should be provided if requested.

- iv. **Independent Assessment** - The Contractor (and/or subcontractor) must have an independent third party validate the security and privacy controls in place for the system(s) commensurate with the risk levels per NIST SP 800-53B. The independent third party must review and analyze the Security Authorization package, and report on technical, operational, and management level deficiencies as outlined in NIST SP 800-53. The Contractor must address all "high" deficiencies *and* enter all other deficiencies that require mitigation by Contractor before submitting the package to

the Government for acceptance and document all remaining deficiencies in a system Plan of Action and Milestones (POA&M).

- v. **POA&M** - as part of the SAR All high-risk weaknesses must be mitigated within 30 calendar days moderate weaknesses must be mitigated within 60 calendar days and low weaknesses must be mitigated within 180 calendar days from the date the weaknesses are formally identified and documented. Identified risks stemming from deficiencies related to the security control baseline implementation, assessment, continuous monitoring, vulnerability scanning, flaws and security defects in a system (that require to create a patch for remediation), and other security reviews and sources, as documented in the SAR, must be documented and tracked by the Contractor for mitigation in the POA&M document consistent with the HHS Standard for Plan of Action and Milestones, FDA POA&M Management Guide, and FDA IS2P policies. Depending on the severity of the risks, FDA may require designated POA&M weaknesses to be remediated before an ATO is issued. Thereafter, continue to remediate weaknesses throughout the contract. The POA&M document must be updated at least **quarterly**.
- vi. **Contingency Plan and Contingency Plan Test** - due during the annual security assessment. The Contingency Plan must be developed in accordance with NIST SP 800-34, *Contingency Planning Guide for Federal Information Systems*, and be consistent with HHS and FDA policies. Upon acceptance by the System Owner, the Contractor, in coordination with the System Owner, must test the Contingency Plan and prepare a Contingency Plan Test Report that includes the test results, lessons learned and any action items that need to be addressed. Thereafter, the Contractor must update and test the Contingency Plan at least **annually**.
- vii. **E-Authentication Questionnaire** - The Contractor (and/or any subcontractor) must collaborate with government personnel to ensure that the E-Authentication requirements are implemented in accordance with OMB 04-04 and NIST SP 800-63 B.

Based on the level of assurance determined by the E-Auth, the Contractor (and/or subcontractor) must ensure appropriate authentication to the system, including remote authentication, is in-place in accordance with the assurance level determined by the E-Auth (when required) in accordance with HHS *Guidance for Selection of e-Authentication Assurance Levels* and any other applicable HHS and FDA policies.

ii. **Information Security Continuous Monitoring**. Upon the government issuance of an Authority to Operate (ATO), the Contractor (and/or subcontractor)-owned/operated systems that input, store, process, output, and/or transmit government information, must meet or exceed the information security continuous monitoring (ISCM) requirements in accordance with FISMA and NIST SP 800-137, *Information Security Continuous Monitoring*

(ISCM) for Federal Information Systems and Organizations, HHS ISCM Strategy, and HHS and FDA IS2Ps.

iii. **Annual Assessment/Penetration (Pen) Test** - Assess the system security and privacy controls (or ensure an assessment of the controls is conducted) at least annually to determine the implemented security and privacy controls are operating as intended and producing the desired results (this involves penetration testing conducted by the FDA. In addition, review all relevant A&A documentation (SSP, POA&M, Contingency Plan, etc.) and provide updates by specified due date in the deliverable table.

iv. **Asset Management** – Using FDA-approved Security Content Automation Protocol (SCAP)-compliant automated tools for active/passive scans, provide an inventory of all information technology (IT) assets for hardware and software, (computers, servers, routers, databases, operating systems, etc.) that are processing FDA-owned information/data. It is anticipated that this inventory information will be required to be produced at least annually to facilitate management/oversight efforts. IT asset inventory information must include IP address, machine name, operating system level, security patch level, and SCAP-compliant format information. The Contractor must maintain a capability to provide an inventory of 100% of its IT assets using SCAP-compliant automated tools in accordance with the *HHS Policy for Information Technology Asset Management (ITAM)* and any other applicable HHS and FDA policies.

v. **Configuration Management** - Use FDA-approved SCAP-compliant automated tools as per NIST IR 7511 and *HHS Minimum Security Configurations Standards Guidance* to scan all IT assets, including but not limited to: computers, servers, routers, databases, operating systems, application, etc., that store and process government information. Provide scan reports to HHS/ FDA upon request. The Contractor must maintain a capability to provide security configuration compliance information for 100% of its IT assets using SCAP-compliant automated tools.

vi. **Vulnerability Management** - Contractors must actively manage system vulnerabilities using automated tools and technologies where practicable and in accordance with the *FDA Information Security and Privacy Protection (IS2P)* policy and *HHS Policy for Vulnerability Management*. Automated tools must be compliant with NIST-specified SCAP standards for vulnerability identification and management. The Contractor must maintain a capability to provide security vulnerability scanning information for 100% of IT assets using SCAP-compliant automated tools and report to the agency at least monthly.

vii. **Patching and Vulnerability Remediation** - Install vendor released security patches and remediate critical and high vulnerabilities in systems processing government information in an expedited manner, within vendor and FDA patch management policy timeframes.

viii. **Secure Coding** - Follow the *HHS Policy for Software Development Secure Coding Practices* and secure coding best practice requirements, as directed by United States Computer Emergency Readiness Team (US-CERT) specified standards and the Open Web Application Security Project (OWASP), that will limit system software vulnerability exploits.

ix. **Boundary Protection** - The Contractor must ensure that government information, other than unrestricted information, being transmitted from federal government entities to external entities is routed through a Trusted Internet Connection (TIC).

c. **Government Access for Security Assessment.** In addition to the Inspection Clause in the contract, the Contractor (and/or any subcontractor) must afford the Government access to the Contractor's facilities, installations, operations, documentation, information systems, and personnel used in performance of this contract to the extent required to carry out a program of security assessment (to include vulnerability testing), investigation, and audit to safeguard against threats and hazards to the confidentiality, integrity, and availability of federal i. At any tier handling or accessing information, consent to and allow the Government, or an independent third party working at the Government's direction, without notice at any time during a weekday during regular business hours Contractor local time, to access Contractor and subcontractor installations, facilities, infrastructure, data centers, equipment (including but not limited to all servers, computing devices, and portable media), operations, documentation (whether in electronic, paper, or other format) The Government includes but is not limited to the U.S. Department of Justice, U.S. Government Accountability Office, and the HHS Office of the Inspector General (OIG). The purpose of the access is to facilitate performance inspections and reviews, security and compliance audits, and law enforcement investigations. For security audits, the audit may include but not be limited to such items as buffer overflows, open ports, unnecessary services, lack of user input filtering, cross site scripting. At any tier handling or accessing protected information, fully cooperate with all audits, inspections, investigations, forensic analysis, or other reviews or requirements needed to carry out requirements presented in applicable law or policy. Beyond providing access, full cooperation also includes, but is not limited to, disclosure to investigators of information sufficient to identify the nature and extent of any criminal or fraudulent activity and the individuals responsible for that activity. It includes timely and complete production of requested data, metadata, information, and records relevant to any inspection, audit, investigation, or review, and making employees of the Contractor available for interview by inspectors, auditors, and investigators upon

request. Full cooperation also includes allowing the Government to make reproductions or copies of information and equipment, including, if necessary, collecting a machine or stem image capture. Segregate Government protected information and metadata on the handling of Government protected information from other information. Commingling of information is prohibited. Inspectors, auditors, and investigators will not be precluded from having access to the sought information if sought information is commingled with other information. Cooperate with inspections, audits, investigations, and reviews.

d. End of Life Compliance. The Contractor (and/or any subcontractor) must use Commercial off the Shelf (COTS) software or other software that is supported by the manufacturer. In addition, the COTS/other software need to be within one major version of the current version; deviation from this requirement will only be allowed via the HHS waiver process (approved by HHS CISO if it impacts enterprise-wide systems and services, or by the FDA CISO if it impacts only the FDA). The Contractor must retire and/or upgrade all software/systems that have reached end-of-life in accordance with *HHS End of Life Operating Systems, Software and Application Policy* and *FDA End-of-Life Operating Systems, Software, and Applications Policy*.

e. Desktops, Laptops, and Other Computing Devices Required for Use by the Contractor. The Contractor (and/or any subcontractor) must ensure that all IT equipment (e.g., laptops, desktops, servers, routers, mobile devices, peripheral devices, etc.) used to process information on behalf of FDA are deployed and operated in accordance with approved security configurations and meet the following minimum requirements:

- i. Encrypt equipment and sensitive information stored and/or processed by such equipment in accordance with HHS and FDA encryption standard and current FIPS 140 validation certificates from the NIST CMVP.
- ii. Configure laptops and desktops in accordance with the latest applicable United States Government Configuration Baseline (USGCB), FDA Configuration Baselines, FDA Minimum Security Configuration Standards, and HHS Minimum Security Configuration Standards;
- iii. Maintain the latest operating system patch release and anti-virus software definitions, per FDA patch management policy;
- iv. Validate the configuration settings after hardware and software installation, operation, maintenance, update, and patching and ensure changes in hardware and software do not alter the approved configuration settings; and
- v. Automate configuration settings and configuration management in accordance with HHS and FDA security policies, including but not limited to:

1. Configuring its systems to allow for periodic Federal, HHS, and FDA vulnerability and security configuration assessment scanning; and
2. Using FDA-approved Security Content Automation Protocol (SCAP)-validated tools with capabilities to scan its systems at least on a monthly basis and report the results of these scans to the CO and/or COR, Project Officer, and any other applicable designated POC.

f. Data Rights

FDA requires the following non-negotiable support across the Enterprise through this contract:

A. Enterprise Portability & Reuse

FDA retains unrestricted rights to export, replicate, and reuse FDA Content and Derived Data outside the Products across the enterprise without additional license or fees. The Contractor shall not restrict integration with other systems as directed by FDA.

B. Open Formats & APIs

The Contractor shall provide bulk export in open, machine-readable formats (CSV, JSON, Parquet) and maintain documented APIs usable without per-connector fees. FDA may use its own pipelines for integration/export.

- **Data Rights**

In addition to 27.404-1 Unlimited rights data, the Government has **unlimited rights** to all documents/material produced under this IDIQ. All documents and materials, to include the source codes of any software, produced under this contract shall be Government owned and are the property of the Government with all rights and privileges of the ownership/copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the Contracting Officer. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. The right does not abrogate any other Government rights under the applicable Data Rights clause(s).

All data collected by the Contractor or provided to the Contractor in the performance of this contract are the property of the Government. The Government retains all rights to the data used and all derivative works developed by the Contractor. The Contractor agrees that during performance of the contract and for a period of six (6) years after the completion of performance of this contract, the Contractor, including all divisions thereof, and any

affiliate of the Contractor, any joint venture involving the Contractor, any entity into or with which it may subsequently merge or affiliate, or any other successor or assign of the Contractor, shall not:

Supply information or material received from this contract, to the public or to any firm participating in or having a known prospective interest in the subject matter areas for which the sensitive information such as the name or mission of the Government agency/department that provided the data was initially submitted.

HHS MANDATED CONTRACTOR TRAINING

HHS mandates that all prime Contractors and sub-Contractors that are performing work under a government contract regardless of having direct access to the agency's network are required to complete annually the HHS mandated Contractor training courses within three (3) business days after contract award; and any other newly mandated training as required by the FDA within ten (10) business after training notification as follows:

- FDA Information Security Awareness – Contractors without direct access to the FDA network, may use the HHS publicly available training site:
<https://www.hhs.gov/about/agencies/asa/ocio/cybersecurity/security-awareness-training/>
 - Training completed at this site is no tracked, therefore keep the HHS training certificates for your records and provided as a separate deliverable to the CO and/or COR.
 - Regular users should complete the following:
 - [Cybersecurity Awareness Training – Module 1](#)
 - [Cybersecurity Awareness Training – Module 2](#)
 - [Rules of Behavior Overview – Module 3](#)
 - If there is a system administrator on the contract, the Contractors shall complete the HHS <https://www.hhs.gov/sites/default/files/hhs-etc/rbt-it-training/index.html> and provide certificates to the CO and/or COR.
- FDA Records Management – Contractors without access to the FDA network but are on an FDA contract and who are creating records on behalf of FDA shall complete the .pdf version of the FDA annual Records Management training and the certificate of completion, and provide the deliverables to the CO and/or COR.
- Emergency Preparedness – the Administration for Strategic Preparedness and Response (ASPR) requires Contractors without FDA Network access to complete the mandatory training via the following link: Emergency Preparedness Training

- The Contractor Program Manager (PM) shall provide to the FDA COR/PM a training completion report via email within five (5) business days before the required FDA Training deadline. The report shall consist of all active Contractor employee's names per the Roster, name of the training and documenting training certification.

5 ACCESS TO NON-PUBLIC INFORMATION

All contactors in senior leadership positions, involved in epidemiologic analyses, or who otherwise have potential access to non-public information as a result of their collaboration with the FDA are required to sign the Contractor's Commitment to Protect Non-Public Information Agreement (Form FDA 3398). To prevent disclosure of non-public information, the prime Contractor is responsible for identifying all relevant employees and subcontractors with potential access to non-public information and ensuring that all forms are completed.

6 PLACE OF PERFORMANCE

Contractor Site

7 CONTRACTOR PERFORMANCE EVALUATIONS

In accordance with Federal Acquisition Regulation (FAR) 42.15, FDA will complete annual and final Contractor performance evaluations on task orders with an estimated value greater than (>) Simplified Acquisition Threshold (SAT), inclusive of options. Annual evaluations will be prepared to coincide with the anniversary date of the task orders. Additional interim performance evaluations may be prepared at Contracting Officer discretion. Final performance evaluations will be completed upon task order. FDA will utilize the Vendor Performance Assessment Reporting System (CPARS) in order to execute annual and final Contractor performance evaluations. CPARS is a secure Internet website located at <http://www.cpars.csd.disa.mil/cparsmain.htm>. FDA will register the Contractor in CPARS upon receipt of the name and email address of the individual who will be responsible for serving as the Contractor's CPARS contacts. Once FDA registers the vendor in CPARS, the Contractor will receive an automated CPARS email message which contains User ID and instructions for creating a password. Once a performance evaluation is issued, the Contractor's primary and alternate CPARS contact will receive an email instructing them to logon to CPARS in order to review the performance evaluation. The Contractor has 14 days from the date of performance evaluation issuance in which to review the evaluation. If the Contractor agrees with the performance evaluation outcome, the evaluation becomes final. Should the Contractor disagree with the performance evaluation outcome, rebuttal comments must be submitted via the CPARS within 14 days from date the evaluation was issued by FDA. Any disagreement between the Contracting Officer and the Contractor will be referred to

an acting official one level above the Contracting Officer, whose decision will be final. Copies of each performance evaluation and Contractor responses, if any, will be retained as part of the official contract file and will be used to support future award decisions. Evaluations will also be stored for a 3-year period in the CPARS system. Contractors may obtain CPARS training material and register for on-line training at <http://www.cpars.csd.disa.mil/allapps/cpcbtdlf.htm>. There is no fee for registration or use of the CPARS.

8 OWNERSHIP OF WORK PRODUCTS

All work products created and delivered under this contract (e.g., workflow processes, SOPs, graphics software, software tools, and artifacts) shall be the sole property of FDA. If the Contractor wishes to use any of the information or products created during the course of this contract for purposes other than for this contract, it shall obtain the written permission of FDA.

9 MONTHLY PROGRESS AND FINANCIAL REPORTING FOR LABOR-HOUR TASK ORDERS

Monthly Progress and Financial reports shall be submitted by the vendor with the invoices for labor-hour task orders in accordance with the expenditure categories as follows:

- a. Labor hours and total labor charges for each skill level category for the current period and cumulative cost for the total contract to date.
- b. Cumulative total for all cost categories (i.e., labor).
- c. Task Order estimated value and percent expended to date.
- d. Estimate to complete.

10 ATTACHMENT LIST OF APPLICABLE DOCUMENTS

The Contractor shall perform all services essential for the performance of the contract/task order per each of the attachments listed below.

Attachment	Title
Attachment H	Data Characteristics Table

11 GLOSSARY OF ABBREVIATIONS, ACRONYMS AND DEFINITIONS

Acronym/Term	Description
Adjudicated	Adjudicated claims data is information gathered and analyzed during the process of evaluating and processing insurance claims. This data includes details such as the types of services or goods claimed, amounts, and dates. Claims adjudication is the process by which insurance companies review healthcare claims to determine whether to pay them in full, in part, or deny them.
AOT	The Acquisition Operations Team (AOT) within the Regulatory Science Staff (RSS) is responsible for the oversight and management of the OSE Acquisition and Budget Portfolio. AOT provides acquisition and project management life cycle support to OSE's Regulatory Science Programs: Drug Utilization, Sentinel, Regulatory Information Systems and Regulatory Science and Applied Research.
AQL	Acceptable Quality Level – the minimum percent of deliverables which are compliant with a given performance standard.
ARIA	Active Risk Identification and Analysis System - The FDA Amendments Act of 2007 (FDAAA) Section 905 required that FDA strengthen its existing post-market safety surveillance for drugs and biological products by directing FDA to create an Active post-market risk identification and analysis system. FDA fulfills the Congressional Mandate for a post-market risk identification and analysis system through the Active Risk Identification and Analysis (ARIA) System, a sub-component within the Sentinel 3.0 System. FDAAA Section 901 required that FDA determine whether the post-market risk identification and analysis system is sufficient to identify or assess a serious risk prior to requiring post-market studies or clinical trials.
ARIA Closure Memos	The closure memo is drafted when a Sentinel 3.0 study has been completed and is an internal document and not publicly posted.
ARIA Sponsor Notification Letters	Notifies an applicant or a holder of a new drug application or a biologics license application about certain types of safety studies conducted in the Sentinel 3.0 System.
CBOC	The Community Building and Outreach Center in Sentinel 3.0 engaged with non-FDA participants, from the clinical research enterprise to health advocacy groups, to broaden and activate a strong community to advance Sentinel's objectives.
CDER	The Center for Drug Evaluation and Research (CDER) performs an essential public health task by making sure that safe and effective drugs are

Acronym/Term	Description
	available to improve the health of people in the United States. As part of the U.S. Food and Drug Administration (FDA), CDER regulates over the counter and prescription drugs, including biological therapeutics and generic drugs.
CDM	Common Data Model: A standard data structure for health data (e.g., claims data, EHR data) that allows data to be shared between applications and data sources.
CMS	Centers for Medicare and Medicaid Services
CO	Contracting Officer ensure performance of all necessary actions for effective contracting, ensure compliance with the contract terms, and shall safeguard the interests of the United States in the contractual relationship. The CO shall also ensure that the Contractor receives impartial, fair, and equitable treatment under this contract. Determine the final assessment of the Contractor's performance.
COR	Contracting Officer's Representative (COR) is responsible for providing continuous technical oversight of the Contractor's performance. The COR is not empowered to make any contractual commitments or to authorize any contractual change on the Government's behalf.
COVID-19	Coronavirus disease is an infectious disease caused by the SARS-CoV-2 virus with the first known case in humans detected in 2019.
Data Curation	The process of organizing, cleaning, and preserving data so it can be used.
Data Lag	Claims data lag is the time between when a service is performed and when the data is available for analysis. The length of the lag can vary depending on the type of service and program.
Data Quality Assessment Plan	A structured document outlining the process for evaluating the accuracy, completeness, consistency, timeliness, and relevance of a dataset, identifying potential issues, and proposing corrective actions to ensure the data is reliable for intended use; it typically includes defining data quality objectives, identifying key data elements, selecting assessment methods, and establishing reporting procedures to monitor and improve data quality over time.
Data Refresh	Claims data refresh is the process of updating or replacing existing claims data with the most recent data available. This process is important for ensuring that the information used for analysis, reporting, and decision-making is accurate and up to date
Descriptive Studies	Describes the distribution within a population based on factors like "person, place, and time."

Acronym/Term	Description
DPN	Data Partners Network (DPN): These were the health care organizations, known as Data Partners under Sentinel 2.0, that ran analyses through a data program and sent the deidentified results back to the Sentinel 2.0 coordinating center.
EHR	Electronic Health Records are digital versions of a patient's medical records (or portions of a patient's medical records) that can be accessed by authorized healthcare providers. EHRs can include a variety of information, such as: Diagnoses, Medications, Allergies, Immunizations, Treatment plans, Progress notes, Vital signs, Laboratory data, and Radiology reports. However, a vendor with access to EHR data, may only have data on select variables (such as structured data on select labs) and may or may not have access to the underlying medical record or be able to trace the variable to the underlying source data.
Epi Team	Epidemiology Team - The Epidemiology Team provides pharmacoepidemiology subject matter expertise and serves as liaison between the internal FDA Sentinel 3.0 participants, and Contractors supporting the Sentinel 3.0 System.
FDA	Food and Drug Administration - The Food and Drug Administration (FDA) is responsible for protecting the public health by assuring the safety, efficacy, and security of human and veterinary drugs, biological products, medical devices, our nation's food supply, cosmetics, and products that emit radiation.
FDAAA	The FDA Amendments Act of 2007
Fit-for-Purpose	Data are relevant, valid, and reliable for the intended use (supporting an analysis or study) and whether the provenance and quality of data are trusted by decision-makers
Health Insurance Claims Data	Information about patient interactions with healthcare providers and insurers that is generated when healthcare providers request payment from a health plan. It's a valuable source of healthcare information that's used by insurance companies to process claims and determine who should be paid. The data can also be used in pharmacoepidemiology studies.
Health Insurance Claims Aggregator	The vendor aggregates claims and enrollment data into databases from large health plans linking paid claims to detailed patient information across sites, providers, and time.
HHS	United States Department of Health and Human Services
Health Insurance	The Health Insurance Portability and Accountability Act (HIPAA) of 1996 (Public Law No. 104-191) is a federal law that governs: 1) health insurance

Acronym/Term	Description
Portability and Accountability Act (HIPAA)	coverage for workers and their families when they change or lose their jobs; and 2) the establishment of national standards for electronic healthcare data transactions. https://www.gpo.gov/fdsys/pkg/FR-2017-01-19/pdf/2017-01058.pdf https://www.sentinelinitiative.org/sites/default/files/communications/publications-presentations/HIPPA-Common-Rule-Compliance-in-Sentinel-Initiative.pdf
HIPAA Privacy Rule	The HIPAA Privacy Rule (45 CFR Part 160 and Subparts A and E of Part 164) establishes national standards to protect individuals' medical records and other individual health information and applies to health plans, healthcare clearinghouses, and those healthcare providers that conduct certain healthcare transactions electronically. The Rule requires appropriate safeguards to protect the privacy of individual health information and sets limits and conditions on the uses and disclosures that may be made of such information without patient authorization. https://www.gpo.gov/fdsys/pkg/FR-2017-01-19/pdf/2017-01058.pdf
HIE	Health Information Exchange: The mobilization of health care information electronically across organizations within a region, community, or hospital system. Participants in data exchange are called in the aggregate Health Information Networks (HIN). In practice, the term HIE may also refer to the health information organization (HIO) that facilitates the exchange. The goal of HIE is to facilitate access to and retrieval of clinical data to provide to public health authorities in analyses of the health of the population.
HOI	Health Outcomes of Interest
IAA	Inter-Agency Agreement
IC	The Innovation Center (IC) in Sentinel 3.0 focused on meeting the Real-World Evidence (RWE) Medical Data Enterprise requirement by establishing a query-ready distributed data network containing EHRs of at least 10 million lives, developing and applying innovative analytic methods (e.g., causal inference, feature engineering).
Inferential Studies	Tests hypotheses and make inferences about causal relationships between exposures and outcomes by analyzing data and using approaches to control for other variables that may increase or decrease the estimated association.
IRB	Institutional Review Board
Medical Record	A "medical record" and a "medical chart" are essentially the same thing, referring to a comprehensive document that details a patient's health history, including diagnoses, treatments, medications, allergies, and test results, providing healthcare providers with all necessary information to

Acronym/Term	Description
	make informed decisions about their care; the terms are often used interchangeably. (Electronic Health Record data is not synonymous with medical chart.)
Mini-Sentinel	The Sentinel 3.0 Initiative began with the Mini-Sentinel 3.0 pilot in 2009. The Mini-Sentinel 3.0 pilot created a format for the data and a process for analyzing it. This common data model and distributed data approach enabled the FDA to track the performance of medical products and safeguard patient privacy. The Mini-Sentinel 3.0 pilot was completed in 2014.
MIL	Mother-Infant linked records: In order to evaluate the effect of exposures in pregnancy on outcomes in infants, it is necessary to link mothers/deliveries with infants.
ML	Machine Learning
NLP	Natural Language Processing
OMOP	The Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) is an open community data standard, designed to standardize the structure and content of observational data and to enable efficient analyses that can produce reliable evidence. A central component of the OMOP CDM is the OHDSI standardized vocabularies. The OHDSI vocabularies allow organization and standardization of medical terms to be used across the various clinical domains of the OMOP common data model and enable standardized analytics that leverage the knowledge base when constructing exposure and outcome phenotypes and other features within characterization, population-level effect estimation, and patient-level prediction studies. ,
OSE	Office of Surveillance and Epidemiology - The Office of Surveillance and Epidemiology (OSE) within the Center for Drug Evaluation and Research (CDER) works to detect, assess, prevent, and manage the risks of medications so that they can be relied upon to treat disease and improve health. OSE participates in the safety analysis of drugs before they are marketed to patients and consumers. After the drugs are marketed, we utilize risk assessment tools to identify and assess adverse events and medication errors that did not appear during the drug development process as well as to understand better those risks observed in clinical trials. OSE also provides expertise on risk mitigation strategies to prevent medication errors, to improve the safe use of products, and to manage the risks associated with medicines after they are marketed.
Patient Profiles	A "claims data patient profile" refers to a collection of information about a patient compiled from healthcare claims submitted by providers to

Acronym/Term	Description
	insurance companies, essentially creating a longitudinal record of a patient's medical history, including diagnoses, procedures, medications, and provider visits, based on the data captured in those claims whenever they seek care. Patient profiles can be generated from structured insurance claims data to help understand the patients' longitudinal medical profile. For examples/references, see Sub-Task B: Data Analytics and Reporting.
PM	Program Manager - serves as the Contractor's single contract manager and shall be the authorized interface with the contract level Contracting Officer's Representative (COR), government management personnel, and customer agency representatives.
PMT	Sentinel 3.0 Program Management Team - The PMT is responsible for all operational aspects of the day-to-day activities of the Sentinel 3.0 System. The PMT ensures projects are completed on time and in accordance with FDA regulations; collaborates with participants to facilitate coordination and project completion; works with participants to identify areas of improvement and develop solutions; and identifies any potential risks to projects or deliverables and develop plans to mitigate them. In addition, the PMT facilitates all internal and external communications while abiding by the established FDA's guidance and regulations.
Project Manager	Project Manager - Serves as the project manager for a large, complex health IT task order (or a group of task orders affecting the same common/standard/migration system) and shall assist the Program Manager in working with the contract-level COR, the task order-level COR(s), government management personnel and customer agency representatives. Under the guidance of the Program Manager, responsible for the overall management of the specific task order(s), and manages, plans and coordinates activities of Health IT projects ensuring that the technical solutions and schedules in the task order are implemented in a timely manner.
Protocol-based Assessment	This refers to conducting a study through use of a protocol -- a structured and standardized plan outlining the methods and procedures to be followed when conducting an epidemiological study, ensuring consistency and quality in data collection and analysis, including details on study population selection, exposure and outcome measurements, data collection tools, and analysis techniques. A detailed roadmap for the study.
PWS	Performance Work Statement
QA/QC	Quality Assurance: the maintenance of a desired level of quality in a service or product, especially by means of attention to every stage of the

Acronym/Term	Description
	process of delivery or production. Quality Control: a system of maintaining standards in manufactured products by testing a sample of the output against the specification.
Registry	A place to store detailed information about people with a specific disease or condition, who provide it on a voluntary basis.
Registries (Network of)	A group of registries (e.g., product- or disease-based registries) that work together to share information, establish best practices, or conduct research
RSS	Regulatory Science Staff - RSS manages drug safety surveillance and research programs that are at the forefront of post-marketing safety evaluation and research. Through these programs, RSS leads the identification, evaluation and development of new data sources, tools, and methods to assess product safety. To support the scientific operations, RSS manages OSE extramural budget and acquisitions of data sources, tools, and associated services throughout their acquisitions' lifecycle.
RWD	Real World Data - Real-world data are data relating to patient health status and/or the delivery of health care routinely collected from a variety of sources.
Regulatory Operations	Responses to Congressional inquiries, user-fee commitments, task, and activities directly associated to regulatory requirements such as sponsor notification letters, Sentinel 3.0 performance metrics tracking, PDUFA mandated Sentinel 3.0 Assessment Reports, ARIA sufficiency documentation, public posting of Sentinel 3.0 study results and public presentations.
RWE	Real World Evidence - Real-world evidence is the clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of RWD.
SAP	Statistical Analysis Plan
SCDM	Sentinel 3.0 Common Data Model: A standard data structure for claims data that allows Data Partners to quickly execute distributed programs against local data.
SDD	Sentinel 3.0 Distributed Database
Sequential Testing	In this RFP, sequential testing is a method of data analysis where researchers continuously monitor incoming data in near real-time, performing statistical tests at regular intervals to quickly identify potential disease outbreaks or safety signals, allowing for rapid response and intervention as new information becomes available.

Acronym/Term	Description
Signal Identification Study	In this RFP, a signal identification study refers to a method that uses a statistical technique to identify potential signals or patterns within large datasets, particularly in healthcare settings, by analyzing hierarchical data structures like medical diagnosis codes, effectively detecting unusual trends or potential adverse drug events within a complex system of related outcomes.
SOC	The Sentinel 3.0 Operations Center (SOC) served as the coordinated center in Sentinel 3.0 and provided oversight to the ARIA System, further developed the infrastructure, and maintained the distributed data network.
SOP	Standard Operating Procedures - A standard operating procedure is a set of step-by-step instructions for performing a routine activity. SOPs should be followed the same way every time to guarantee that the organization remains consistent and in compliance with regulations and business standards.